

Beautiframe

Beautiful Theorem-Like Environments for Typst

Version 0.1.0

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1 Introduction

Beautiframe is a Typst package for creating beautiful theorem-like environments (theorems, definitions, lemmas, proofs, etc.) with multiple visual styles.

1.1 Features

- **7 distinct styles:** classic, modern, elegant, colorful, boxed, minimal, academic
- **6 variants per style:** prominent, standard, subtle, accent, minimal, inline
- **Flexible mapping:** Assign any variant to any environment type
- **Independent counters:** Each environment type has its own counter
- **Customizable labels:** Change “Theorem” to “Théorème”, “Satz”, etc.
- **QED symbol presets:** □, ■, ■, CQFD,
- **Color themes:** Pre-built themes (ocean, forest, sunset, lavender)
- **Language presets:** French, German, Spanish
- **Print-friendly modes:** color, grayscale, black & white

1.2 Quick Start

```
#import "@local/beautiframe:0.1.0": *

#theorem(name: "Pythagorean")[
  In a right triangle:  $a^2 + b^2 = c^2$ 
]

#definition[
  A *limit* is the value that a function approaches.
]

#proof[
  The proof is left as an exercise.
]
```

2 Environments

2.1 Available Environments

Beautiframe provides 8 environment types:

Environment	Default Variant	Counter	Usage
theorem	prominent	Optional	Main results
definition	standard	Optional	Foundational concepts
lemma	standard	Optional	Supporting results
proposition	standard	Optional	Secondary results
corollary	standard	Optional	Consequences
remark	subtle	Optional	Commentary
example	accent	Optional	Illustrations
proof	(special)	No	Demonstrations

2.2 Basic Usage

Each environment function accepts:

- **name:** Optional name (e.g., “Pythagorean”)
- **number:** auto (default), none, or custom value
- **body:** The content

```
#theorem(name: "Fermat's Last") [
  There are no positive integers $a$, $b$, $c$ such that
  $a^n + b^n = c^n$ for $n > 2$.
]
```

Theorem 1 | There are no positive integers a , b , c such that $a^n + b^n = c^n$ for $n > 2$.
Fermat's Last

2.3 Numbering Control

2.3.1 Automatic Numbering (default)

```
#theorem[First theorem] // Theorem 1
#theorem[Second theorem] // Theorem 2
```

2.3.2 No Numbering

```
#theorem(number: none)[Unnumbered theorem]
#definition(number: none)[Unnumbered definition]
```

Theorem | This theorem has no number.
Example

2.3.3 Custom Numbering

```
#theorem(number: "A")[Special theorem A]
#theorem(number: "★")[Starred theorem]
```

Theorem A | Special theorem A

2.3.4 Numbered Remarks

By default, remarks have no number. To number them:

```
#remark(number: auto)[This remark will be numbered.]
```

Remark 1 | This remark is numbered.

Remark | This remark is not numbered (default).

2.3.5 Counter Reset

Reset all counters manually:

```
#beautiframe-reset()
```

3 Styles

Beautiframed includes 7 visual styles. Each style provides 6 variants.

3.1 Classic Style

Traditional textbook layout with badge on the left and content with left border. Based on the exercise-bank pattern.

```
#beautiframed-setup(style: "classic")
#theorem(name: "Classic")[In a right triangle:  $a^2 + b^2 = c^2$ ]
#definition[A continuous function preserves limits.]
#remark[This is the default style.]
```

Theorem 1 | In a right triangle: $a^2 + b^2 = c^2$
Classic

Definition 1 | A continuous function preserves limits.

Remark | This is the default style.

3.2 Modern Style

Contemporary design with thick accent bars and rule separators.

```
#beautiframed-setup(style: "modern")
#theorem(name: "Modern")[In a right triangle:  $a^2 + b^2 = c^2$ ]
#definition[A continuous function preserves limits.]
```

Theorem 1 — *Modern*

In a right triangle: $a^2 + b^2 = c^2$

Definition 1

A continuous function preserves limits.

3.3 Elegant Style

Sophisticated styling with centered headers and decorative ornaments.

```
#beautiframed-setup(style: "elegant")
#theorem(name: "Elegant")[In a right triangle:  $a^2 + b^2 = c^2$ ]
#definition[A continuous function preserves limits.]
```

◇ **THEOREM 1** ◇

(Elegant)

In a right triangle: $a^2 + b^2 = c^2$

• **DEFINITION 1** •

A continuous function preserves limits.

3.4 Colorful Style

Each environment type has a distinct color for visual differentiation.

```
#beautiframed-setup(style: "colorful")
#theorem[Red for theorems.]
#definition[Blue for definitions.]
#example[Green for examples.]
```

Theorem 1

Red for theorems.

Definition 1

Blue for definitions.

Example 1

Green for examples.

3.5 Boxed Style

Full framed boxes with optional backgrounds and header bars.

```
#beautiframed-setup(style: "boxed", border-radius: 3pt)
#theorem(name: "Boxed")[In a right triangle:  $a^2 + b^2 = c^2$ ]
#definition[A continuous function preserves limits.]
```

Theorem 1 (*Boxed*)

In a right triangle: $a^2 + b^2 = c^2$

Definition 1

A continuous function preserves limits.

3.6 Minimal Style

Ultra-clean, print-friendly. Minimal ink usage.

```
#beautiframing-setup(style: "minimal")
#theorem(name: "Minimal")[In a right triangle:  $a^2 + b^2 = c^2$ ]
#definition[A continuous function preserves limits.]
```

Theorem 1 (*Minimal*).

In a right triangle: $a^2 + b^2 = c^2$

Definition 1. A continuous function preserves limits.

3.7 Academic Style

Formal research paper style matching AMS/journal conventions.

```
#beautiframing-setup(style: "academic")
#theorem(name: "Academic")[In a right triangle:  $a^2 + b^2 = c^2$ ]
#definition[A continuous function preserves limits.]
```

THEOREM 1 (*Academic*).

In a right triangle: $a^2 + b^2 = c^2$

Definition 1.

A continuous function preserves limits.

4 Variants

Each style provides variants that can be assigned to any environment type. The 6 core variants are available in all styles:

Variant	Description	Default for
prominent	Strongest visual emphasis, thick borders, bold colors	theorem
standard	Normal, balanced styling	definition, lemma, proposition, corollary
subtle	Lighter, less prominent, muted colors	remark
accent	Uses environment-specific color	example
minimal	Very light styling, minimal visual elements	—
inline	No structural elements, flows with text	—

Boxed style has 3 additional variants:

Variant	Description
titled	Label breaks the top border line (beautitled-style)
centered	Label centered at top, breaking the border
corner	L border (top + left sides)
corner2	Inverted L border (left + bottom sides)

4.1 Assigning Variants

Map any variant to any environment type:

```
#beautifframe-setup(
  theorem-variant: "prominent", // Theorems get maximum emphasis
  lemma-variant:  "subtle",     // Lemmas are de-emphasized
  remark-variant: "inline",     // Remarks flow inline
  example-variant: "accent",    // Examples use their color
)
```

4.2 Visual Gallery: All Variants × All Styles

4.2.1 Classic Style

Prominent:

Theorem 1 | Sample text.
Name

Subtle:

Theorem 3 | Sample text.
Name

Minimal:

Theorem 5 Sample text.
Name

Standard:

Theorem 2 | Sample text.
Name

Accent:

Theorem 4 | Sample text.
Name

Inline:

Theorem 6 (*Name*). Sample text.

4.2.2 Modern Style

Prominent:

Theorem 1 — *Name*

Sample text.

Subtle:

Theorem 3 — *Name*

Sample text.

Minimal:

Theorem 5 — *Name*

Sample text.

Standard:

Theorem 2 — *Name*

Sample text.

Accent:

Theorem 4 — *Name*

Sample text.

Inline:

Theorem 6 — *Name* — Sample text.

4.2.3 Elegant Style

Prominent:

◇ **THEOREM 1** ◇
(Name)

Sample text.

Subtle:

Theorem 3 (Name). Sample text.

Minimal:

THEOREM 5
(Name)

Sample text.

Standard:

• **THEOREM 2** •
(Name)

Sample text.

Accent:

★ **THEOREM 4** ★
(Name)

Sample text.

Inline:

THEOREM 6 *(Name)*. Sample text.

4.2.4 Colorful Style

Prominent:

Theorem 1 (*Name*)

Sample text.

Subtle:

Theorem 3 (*Name*)

Sample text.

Minimal:

Theorem 5 (*Name*)

Sample text.

Standard:

Theorem 2 (*Name*)

Sample text.

Accent:

Theorem 4 (*Name*)

Sample text.

Inline:

Theorem 6 (*Name*). Sample text.

4.2.5 Boxed Style

Prominent:

Theorem 1 (*Name*)
Sample text.

Standard:

Theorem 2 (*Name*)
Sample text.

Subtle:

Theorem 3 (*Name*)
Sample text.

Accent:

Theorem 4 (*Name*)
Sample text.

Minimal:

Theorem 5 (*Name*)
Sample text.

Inline:

[**Theorem 6:** *Name*] Sample text.

Titled:

Theorem 7 (*Name*)
Sample text.

Centered:

Theorem 8 (*Name*)
Sample text.

Corner:

Theorem 9 (*Name*)
Sample text.

Corner2:

Theorem 10 (*Name*)
Sample text.

4.2.6 Minimal Style

Prominent:

Theorem 1 (*Name*).

Sample text.

Subtle:

Theorem 3 (*Name*). Sample text.

Minimal:

Theorem 5. Sample text.

Standard:

Theorem 2 (*Name*). Sample text.

Accent:

THEOREM 4 (*Name*). Sample text.

Inline:

(**Theorem 6:** *Name*) Sample text.

4.2.7 Academic Style

Prominent:

THEOREM 1 (*Name*).

Sample text.

Subtle:

Theorem 3 (*Name*). Sample text.

Minimal:

Theorem 5. Sample text.

Standard:

Theorem 2 (*Name*).

Sample text.

Accent:

THEOREM 4 (*Name*).

Sample text.

Inline:

Theorem 6 (*Name*): Sample text.

5 Proofs and QED Symbols

5.1 Basic Proof

```
#proof[
  By direct calculation, we have  $2^2 = 4$ .
]
```

Proof. By direct calculation, we have $2^2 = 4$. □

5.2 QED Symbol Presets

Beautiframe provides several QED symbol presets:

```
#qed-square()      // □ (default)
#qed-filled()      // ■
#qed-tombstone()   // ■
#qed-cqfd()        // CQFD (French)
#qed-slashes()     // //
#qed-text()        // Q.E.D.
#qed-none()        // (no symbol)
```

Default (□):

Proof. Hollow square.

Tombstone (■):

Proof. Tombstone symbol.

Slashes:

Proof. Double slash.

Filled (■):

□ **Proof.** Filled square. ■

CQFD:

■ **Proof.** French style.

CQFD

Q.E.D.:

// **Proof.** Latin abbreviation.

Q.E.D.

5.3 Custom QED Symbol

```
#beautiframe-setup(qed-symbol: text(fill: green, sym.checkmark))
```

Proof. Custom green checkmark. ✓

6 Colors and Themes

6.1 Global Colors

```
#beautiframed-setup(
  primary-color: rgb("#1a5276"),
  secondary-color: rgb("#7f8c8d"),
  accent-color: rgb("#2980b9"),
)
```

6.2 Per-Environment Colors

Used primarily by the colorful style:

```
#beautiframed-setup(
  theorem-color: rgb("#c0392b"), // Red
  definition-color: rgb("#2980b9"), // Blue
  lemma-color: rgb("#8e44ad"), // Purple
  example-color: rgb("#27ae60"), // Green
  remark-color: rgb("#7f8c8d"), // Gray
)
```

6.3 Color Themes

Pre-built color schemes:

Ocean Theme:

Theorem

Blue tones throughout.

Forest Theme:

Theorem

Green tones throughout.

Sunset Theme:

Theorem

Warm red and orange tones.

Lavender Theme:

Theorem

Purple tones throughout.

6.4 Print-Friendly Modes

For B&W printing:

```
#beautiframe-setup(color-mode: "color")    // Full color (default)
#beautiframe-setup(color-mode: "grayscale") // Grayscale
#beautiframe-setup(color-mode: "bw")       // Pure black and white
```

Color mode:**Theorem**

Full color styling.

Grayscale mode:**Theorem**

Grayscale for B&W printers.

B&W mode:**Theorem**

Pure black and white.

7 Language Presets

7.1 French

`#preset-french()`

Théorème 1 | Dans un triangle rectangle: $a^2 + b^2 = c^2$
Pythagore

Définition 1 | Une fonction continue préserve les limites.

Preuve. Immédiat. □

7.2 German

`#preset-german()`

Satz 1 | In einem rechtwinkligen Dreieck: $a^2 + b^2 = c^2$
Pythagoras

Definition 1 | Eine stetige Funktion erhält Grenzwerte.

Beweis. Offensichtlich. □

7.3 Spanish

`#preset-spanish()`

Teorema 1 | En un triángulo rectángulo: $a^2 + b^2 = c^2$
Pitágoras

Definición 1 | Una función continua preserva límites.

Demostración. Inmediato. □

7.4 Custom Labels

```
#beautiframe-setup(
  theorem-label: "Théorème",
  definition-label: "Définition",
  proof-label: "Démonstration",
)
```

8 Layout Configuration

8.1 Classic Style Layout

The classic style uses a grid layout with configurable dimensions:

```
#beautiframed-setup(
  line-position: 2cm,    // Distance from left to vertical line
  label-extra: 1cm,     // Extension into left margin
  border-width: 1.5pt,  // Line thickness
)
```

8.2 Spacing

```
#beautiframed-setup(
  theorem-above: 1.2em, // Space before theorems
  theorem-below: 1em,   // Space after theorems
  header-gap: 0.4em,    // Gap between label and body
)
```

8.3 Typography

Control label appearance:

```
#beautiframed-setup(
  label-weight: "bold", // "bold", "regular", "semibold", etc.
  label-size: 11pt,     // Size of "Theorem", "Definition", etc.
  name-style: "italic", // Style for theorem names: "italic" or "normal"
)
```

Bold label (default):

Teorema 1 (*Name*)

Sample text.

Regular label:

Teorema 2 (*Name*)

Sample text.

8.4 Boxed Style Options

```
#beautiframed-setup(
  inset: (x: 1em, y: 0.8em), // Padding inside boxes
  border-radius: 4pt,        // Rounded corners
  border-width: 1.5pt,       // Border thickness
)
```

9 Custom Environments

Create your own environment types with independent counters using `new-env`:

```
// Create custom environments
#let conjecture = new-env("Conjecture", base: "theorem")
#let propriete = new-env("Propriété", base: "definition", numbered: false)
#let formule = new-env("Formule", base: "lemma", color: green)
#let axiom = new-env("Axiom", base: "theorem", numbered: true)
```

```
// Use them like built-in environments
#conjecture[Every even number greater than 2 is the sum of two primes.]
#conjecture(name: "Goldbach")[Famous unsolved problem.]
#propriete[A property without number.]
#formule[The quadratic formula:  $x = \frac{-b \pm \sqrt{b^2 - 4a c}}{2a}$ ]
```

Conjecture 1

Every even number greater than 2 is the sum of two primes.

Conjecture 2 (*Goldbach*)

Famous unsolved problem in number theory.

Axiom 1

Two points determine a unique line.

9.1 Parameters

```
#let my-env = new-env(
  "Label",           // Display label (required)
  base: "theorem",   // Inherit style from: theorem, definition, lemma, etc.
  numbered: true,     // Auto-number by default
  color: none,        // Optional custom color
)
```

9.2 Dynamic Labels (Plural Forms)

For environments that need dynamic labels (e.g., singular/plural), create wrapper functions:

```
#let definition(plural: false, ..args, body) = {
  let label = if plural { "Définitions" } else { "Définition" }
  beautiframe-setup(definition-label: label)
  env(type: "definition", ..args, body)
}
```

```
// Usage
#definition[A single definition.]
#definition(plural: true)[Multiple definitions grouped together.]
```

9.3 Resetting Custom Counters

```
#reset-env("Conjecture") // Reset specific custom environment
#beautiframe-reset()      // Reset all built-in counters
```

10 API Reference

10.1 Environment Functions

```
#theorem(name: none, number: auto)[body]
#definition(name: none, number: auto)[body]
#lemma(name: none, number: auto)[body]
#proposition(name: none, number: auto)[body]
#corollary(name: none, number: auto)[body]
#remark(name: none, number: none)[body]
#example(name: none, number: auto)[body]
#proof[body]
```

10.2 Setup Function

```
#beautiframed-setup(
  style: "classic",           // Style selection
  // Variant mapping
  theorem-variant: "prominent",
  definition-variant: "standard",
  lemma-variant: "standard",
  proposition-variant: "standard",
  corollary-variant: "standard",
  remark-variant: "subtle",
  example-variant: "accent",
  // Colors
  primary-color: rgb("#2c3e50"),
  secondary-color: rgb("#7f8c8d"),
  accent-color: rgb("#2980b9"),
  theorem-color: rgb("#c0392b"),
  definition-color: rgb("#2980b9"),
  // ... (more per-environment colors)
  // Typography
  label-size: 11pt,
  label-weight: "bold",
  name-style: "italic",
  // Spacing
  theorem-above: 1em,
  theorem-below: 0.8em,
  header-gap: 0.3em,
  // Layout
  inset: (x: 0.8em, y: 0.6em),
  border-width: 1pt,
  border-radius: 0pt,
  line-position: 2cm,
  label-extra: 1cm,
  // Numbering
  link-to-section: false,
  // Labels
  theorem-label: "Theorem",
  definition-label: "Definition",
  // ... (more labels)
  proof-label: "Proof",
  // QED
  qed-symbol: sym.square.stroked,
  // Advanced
  breakable: true,
  color-mode: "color",
)
```

10.3 Utility Functions

```
#beautiframed-reset() // Reset all counters

// Language presets
#preset-french()
#preset-german()
#preset-spanish()
```



```
// Color themes
#theme-ocean()
#theme-forest()
#theme-sunset()
#theme-lavender()

// QED presets
#qed-square()      // □
#qed-filled()      // ■
#qed-tombstone()   // ▀
#qed-cqfd()        // CQFD
#qed-slashes()     // //
#qed-text()        // Q.E.D.
#qed-none()        // (none)
```